

PHYSIOLOGIC EFFECTS OF DIRECTIONAL CHANGES IN INTERMITTENT EXERCISE IN SOCCER PLAYERS

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ABSTRACT

Dellal, A, Keller, D, Carling, C, Chaouachi, A, Wong, DP, and Chamari, K. Physiologic effects of directional changes in intermittent exercise in soccer players. *J Strength Cond Res* 24(12): 3219–3226, 2010—The aim of the present study was to compare the physiologic impact of intermittent exercise in specific shuttle running (IS), which requires 180° directional changes, and traditional in-line (IL) running. Ten elite male adult soccer players performed different intermittent exercises according to their maximal aerobic velocity ($\dot{V}O_{2\max}$): 30–30 seconds at 100% (30 s of runs at 100% of $\dot{V}O_{2\max}$ alternated with 30-s recovery period), 105%, and 110% of $\dot{V}O_{2\max}$ with active recovery, 15–15 seconds at 105%, 110%, and 115% of $\dot{V}O_{2\max}$, and 10–10 seconds at 110%, 115%, and 120% of $\dot{V}O_{2\max}$ with passive recovery. Each exercise was performed in the IL and IS format in a randomized order. Heart rate (HR) expressed in percentage of HR reserve (HRres), postexercise blood lactate concentration [La], and ratings of perceived exertion (RPE) were recorded. The different 30–30 seconds showed significantly higher HRres responses in IS compared with IL ($p < 0.01$). The [La] and RPE results indicated higher values in IS. In conclusion, the physiologic impact of specific IS is substantially higher than in traditional IL. The changes of direction induce an increase in the anaerobic metabolism solicitation and consequently create different responses compared with traditional IL running. This information can aid coaches in the design of intermittent training programs using classical (IL) or a specific form (IS) of running to induce different physiologic responses.

KEY WORDS intermittent exercise, football, fitness training, anaerobic metabolism, 180°, turning

INTRODUCTION

Analyses of soccer match play have highlighted the intermittent nature of the game, and consequently, the capacity to repeat high-intensity exercise bouts is crucial. Intermittent exercise training is commonly used to recreate the actual demands of match play (2,16). This form of training involves alternating work and recovery periods (using active or passive recovery intervals) with the principal aim of optimizing the players' maximal oxygen uptake ($\dot{V}O_{2\max}$) (3). It allows players to work for longer durations than continuous exercise at the same intensity through reduced lactate accumulation (e.g., 22) because lactate is partly metabolized during recovery periods (1). At a physiologic level, intermittent exercise training provides a simultaneous and a mixed solicitation of the aerobic (23) and anaerobic metabolisms (2) and has been shown to improve the oxidative capacity of enzymes (34) and reaction time (27) while impacting the peripheral component of performance (6).

The physiologic responses of traditional high-intensity intermittent exercise using in-line or straight-forward running (IL) are well-known (e.g. 3,15). However, activity profiles in soccer show that players do not only carry out in-line running actions. Therefore, if intermittent exercise training included directional changes, it would correspond more to the real demands of the game. Yet, the physiologic impact of intermittent shuttle running exercise (IS) is unclear, especially in soccer training, and the physiologic responses may be different from those obtained in IL running. In this context, no studies have compared responses in IS and IL, and a study on the real effects of IS needs to be made for use within soccer training programs. This type of effort is a common feature of soccer match play and training sessions because players frequently undertake running actions in which they accelerate, decelerate, and change direction before re-accelerating (15). These changes in speed and direction influence the musculature involved, thereby affecting energy use, and could result in higher physiologic responses when compared with habitual forward running movements (15,16).

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In this context, the aim of the present study was to investigate the physiologic impact of directional changes through the comparison of 2 types of high-intensity intermittent exercise: a traditional in-line intermittent exercise protocol versus a specific intermittent shuttle exercise protocol in which 180° directional changes, deceleration, and re-acceleration movements are required. This information can aid coaches in the design of intermittent training exercises programs that induce different training responses using classical (IL) or a specific form (IS) of intermittent exercise. We hypothesized that in-shuttle exercise induces higher physiologic responses than the same running speed in intermittent exercise performed in-line.

METHODS

Experimental Approach to the Problem

The subjects performed different intermittent exercises based on in-line running and shuttle running with 180° directional changes (IS) at several running intensities (Table 1) determined according to their maximal aerobic velocity ($\dot{V}O_{2\max}$). This variable is defined as the lowest velocity that elicits $\dot{V}O_{2\max}$ and is a sensitive measure of aerobic capacity frequently used for prescription of training intensities (4). The $\dot{V}O_{2\max}$ values were obtained using the Leger-Boucher field test (26). Subjects were allowed to take part in testing if they presented no signs of illness/injury or fatigue; otherwise, exercise was deferred to the following day. All sessions were separated by at least 48 hours to minimize the effects of fatigue, and conditions were standardized. All testing sessions were performed on an ash running track or a synthetic soccer pitch to guarantee a controlled surface condition. This choice ensured more stable environmental conditions because the track in the present study was indoors (controlled temperature and humidity). The chronologic organization of testing (Table 2) involved 2 randomized blocks (block 1 and 3) with the measurement of $\dot{V}O_{2\max}$ and a 5-week period in which the subjects performed different IS and IL. These 2 blocks were separated by a period

in which players performed their usual soccer training without any specific fitness program (block 2) to limit the effects of habituation to the intermittent exercises (15,16). Measures of heart rate (HR), blood lactate ([La]), and rating of perceived exertion (RPE) were used to compare the physiologic responses between the 2 exercise types (i.e., IL and IS).

Subjects

Ten high-level soccer players participating in the amateur national championship volunteered to take part in the present study. The anthropometric and fitness characteristics of the subjects are presented in Table 3. The study was conducted according to the Declaration of Helsinki, and the protocol was fully approved by the clinical research ethics committee before the start of the assessments. Written informed consent was received from all players after a detailed explanation about the aims, benefits, and risks involved with this investigation. Players were told they were free to withdraw from the study at any time without penalty.

High-Intensity Intermittent Exercise

After a standardized warm-up that included jogging and dynamic stretching exercises, all subjects performed each intermittent exercise in the 2 experimental conditions: in-line (IL) and shuttle running with 180° directional changes (IS) in a randomized order. All training sessions were performed at the same time of the day to limit the effects of the circadian variations on the measured variables, particularly on HR measures (17). Furthermore, concerning the IS, the calculation of the distance (42 m, 30 m, 21 m) allowed the researchers to define an identical number of directional changes during all shuttle training sessions for all subjects, taking into consideration the different intensities, the different duration periods, and the subjects' $\dot{V}O_{2\max}$.

The intensity of each intermittent exercise was performed in an equal or a higher value than 100% of the individual $\dot{V}O_{2\max}$, with the aim of reaching a consistent time spent at a high percentage of $\dot{V}O_{2\max}$ (32). Players performed

TABLE 1. Short-duration intermittent exercise characteristics.

Intermittent exercise	Intensity (% of $\dot{V}O_{2\max}$)	Type of recovery	Total duration (min)	No. work periods	No. recovery periods	Distance of go-back for intermittent exercise in shuttle, IS (m)	No. 180° directional changes
30–30	100%, 105%, and 110%	Active	1 * 11'30	12	11	42	3
15–15	105%, 110%, and 115%	Passive	1 * 9'45	20	19	30	2
10–10	110%, 115%, and 120%	Passive	1 * 6'50	21	20	21	2

*x-y = intermittent exercise with x work period and y recovery period; $\dot{V}O_{2\max}$ = speed associated with maximal oxygen uptake.

TABLE 2. Experimental set-up.*

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Block 1							
Week 1		Leger-Boucher test N° 1		IS or IL		Match	
Weeks 2–5		IS or IL		IS or IL		Match	
Block 2							
Weeks 6 and 7			No testing				
Block 3							
Week 8		Leger-Boucher test N° 2		IS or IL		Match	
Weeks 9–12		IS or IL		IS or IL		Match	

*IS = shuttle intermittent exercise with 180° directional changes; IL = in-line intermittent exercise.

3 30-30-second intermittent exercise bouts composed of 30-second exercise periods (at 100%, 105%, and 110% of $\dot{V}O_{2\max}$, respectively) followed by 30 seconds of active recovery periods (40% of $\dot{V}O_{2\max}$), in accordance with Billat et al. (5). Subjects also performed 15-15 seconds with passive recovery at 105%, 110%, and 115% of $\dot{V}O_{2\max}$ and 10-10 seconds at 110%, 115%, and 120% of $\dot{V}O_{2\max}$, in accordance with Dupont et al. (19).

Heart Rate

To obtain mean HRreserve percentage (%HRres), the subjects' HR was monitored continuously (5-s intervals) using portable HR monitors (Polar S-810, Polar-Electro, Kempele, Finland) during each exercise. The %HRres allows an interindividual comparison and was calculated using the formula proposed by Karvonen et al. (25): %HRres = (mean exercise HR - resting HR)/(HRmax - resting HR) * 100. The HRmax was considered to be the highest HR value recorded at the end of the Leger-Boucher test. Resting HR was the

minimal value of HR obtained for 3 consecutive interval times after 10 minutes when players were in a quiet room on a mat in the supine position, with their eyes closed and without having performed any prior exercise.

Postexercise Blood Lactate Concentration

Taoutaou et al. (37) previously reported that the postexercise peak [La] was attained at approximately 3 minutes post-exercise when no active recovery was performed. Blood lactate samples were therefore collected at the third minute postexercise from the fingertip after cleaning it by alcohol. The validity of the used portable analyzer (Lactate Pro, Arkray, Japan) has been demonstrated (35).

Borg's Post-Effort Perception Scale

The Borg scale method is based on a subject's RPE and is used to subjectively gauge the subject's level of intensity in testing, training, and competition. Borg (8) reported that the values of RPE can be used to compare physiologic ratings of

TABLE 3. Characteristics of subjects.*

Parameters	Subjects (<i>n</i> = 10) (mean ± <i>SD</i>)
Age (yr)	23.8 ± 2.9
Body mass (kg)	75.5 ± 6.2
Body height (cm)	180.0 ± 7.1
Léger-Boucher test no.1	17.1 ± 0.8
$\dot{V}O_{2\max}$ (km·h ⁻¹)	
Léger-Boucher test no.2	17.2 ± 0.6
$\dot{V}O_{2\max}$ (km·h ⁻¹)	
Maximal HR (beat·min ⁻¹)	191.0 ± 7.1
Resting HR (beat·min ⁻¹)	56.7 ± 6.3
HRres (beat·min ⁻¹)	120.6 ± 43.4

* $\dot{V}O_{2\max}$ = speed associated with maximal oxygen uptake; HR = heart rate.

TABLE 4. Adapted Borg's rate of perceived exertion scale as described by Foster et al. (24).

Estimation	Description
0	Rest
1	Very, very easy
2	Easy
3	Moderate
4	Somewhat hard
5	Hard
6	*
7	Very hard
8	*
9	*
10	Maximal

TABLE 5. Reliability of measurements in study.*

	Intermittent exercises in-line (IL)			Intermittent exercises in-shuttle (IS)		
	ICC	SEM	Effect size	ICC	SEM	Effect size
%HRres	0.97	1.78	0.76	0.98	1.87	0.57
[La]	0.91	0.63	0.62	0.93	0.74	0.61
RPE	0.78	0.33	0.61	0.71	0.15	0.85

*%HRres = percentage of heart rate reserve; [La] = blood lactate concentration; RPE = rate of perceived exertion; ICC = intraclass correlation coefficient.

perceived measurements such as HR or oxygen uptake ($\dot{V}O_2$). In the present study, each subject provided the RPE value corresponding to a subjective appreciation of the effort performed. The RPE scale proposed by Foster et al. (21) was used (Table 4) because it permits the analysis of the global internal load in soccer training and the supramaximal intermittent exercise (14,24). Each estimation was recorded 2 minutes after the effort for each subject by the standardized question, "How was, and how did you feel the exercise," used in previous study (30). No information was provided on previously recorded RPE values obtained in the other intermittent exercise testing. The RPE method was used 5 weeks before the experiment in normal training to ensure that subjects were accustomed to this method.

Statistical Analyses

All values are expressed as means $\pm$ SD. The normality distribution of the data was checked using the Kolmogorov-Smirnov test, and the statistical analysis variance homogeneity was provided by the Hartley test. After confirming normal distribution, a Student's paired *t*-test was used to analyze %HRres, postexercise [La], and posteffort RPE responses to the 2 intermittent exercise protocols at similar intensities. A one-way analysis of variance with repeated measures was also used to compare values for %HRres, [La], and RPE obtained in the 2 forms of intermittent exercises. Statistical significance was set at $p \leq 0.05$. The reliability of each test was assessed by intraclass correlation coefficient and SEM as suggested by Weir (39).

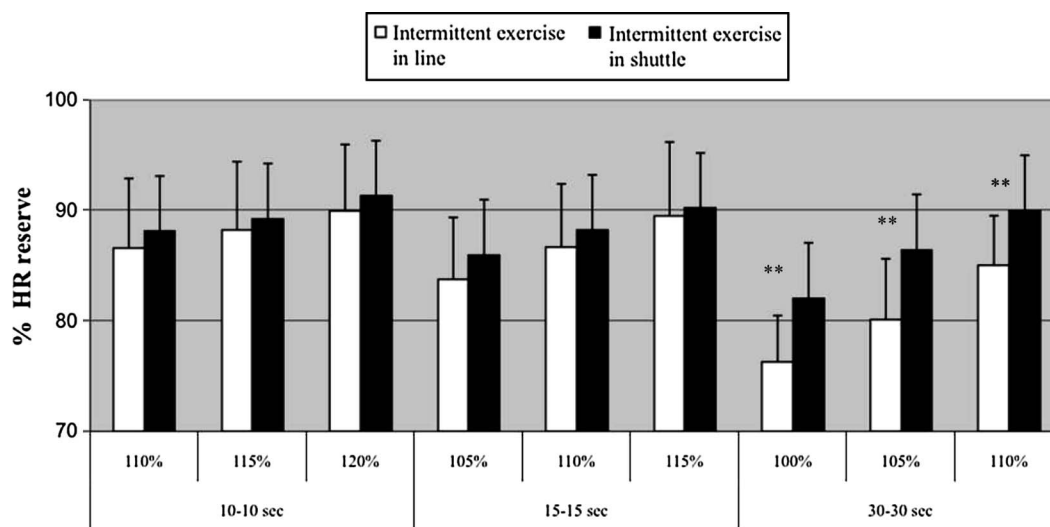


Figure 1. Comparison of percent heart rate reserve response during various intermittent exercises: in-line and shuttle (IL vs. IS). x-y sec: Intermittent exercise with x work period and y recovery period. x%: Percentage of running intensity according to $\dot{V}O_{2\max}$. *Significant different between IL and IS at $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

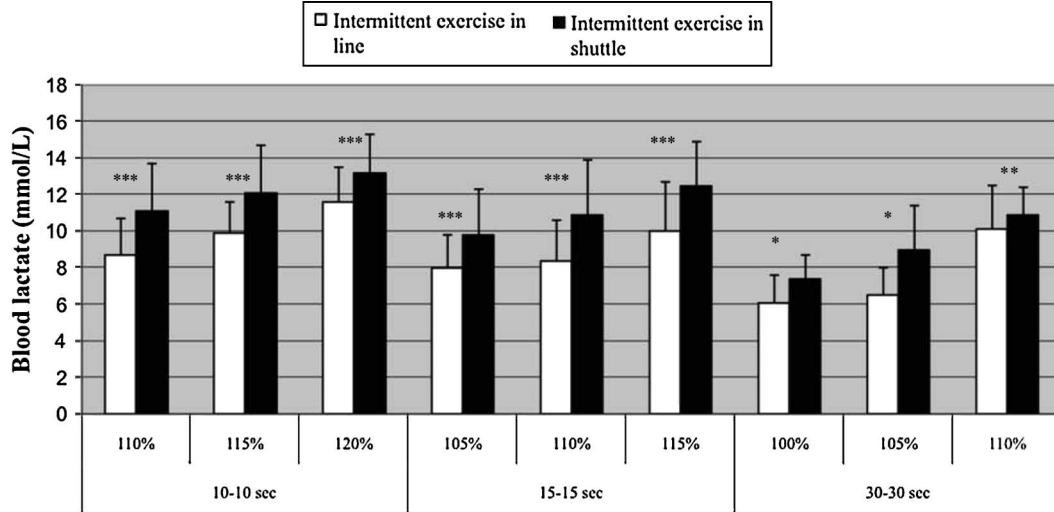


Figure 2. Comparison of postexercise blood lactate responses during various intermittent exercises: in-line and shuttle (IL vs. IS). x-y sec: Intermittent exercise with x work period and y recovery period. x%: Percentage of running intensity according to $\dot{V}O_{2\max}$. *Significant different between IL and IS at $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

RESULTS

All the participants took part in the entire protocol and coped well with the exercises. High reliability was observed in %HRres, [La], and RPE measurements (Table 5). The HR response analysis showed significantly greater ($p \leq 0.01$) values during 30–30-second intermittent exercise performed in shuttle (IS) in comparison with the IL (Figure 1). In contrast, similar HR responses were observed in the 15–15

seconds and 10–10 seconds for both forms of exercise (Figure 1). The comparison of the 2 forms of exercise (IS/IL) at the same intensity demonstrated significant higher [La] values ($p < 0.05$) during IS exercise (Figure 2), especially in the 15–15-second and the 10–10-second protocols ($p < 0.001$).

All RPE values were significantly higher ($p < 0.05$) during IS compared with IL (Figure 3). Maximal values were

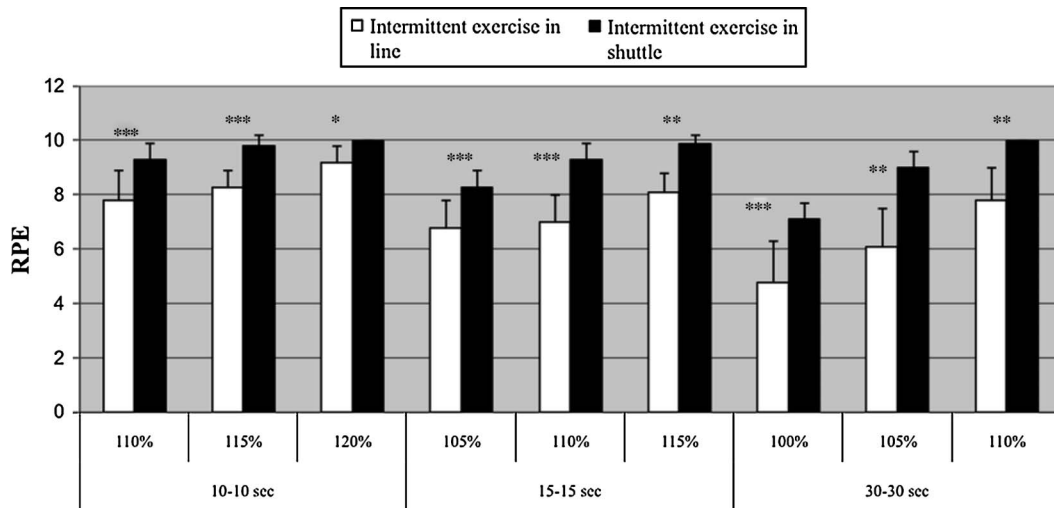


Figure 3. Comparison of rate of perceived effort during various intermittent exercises: in-line and shuttle (IL vs. IS). x-y sec: Intermittent exercise with x work period and y recovery period. x%: Percentage of running intensity according to $\dot{V}O_{2\max}$. *Significant different between IL and IS at $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

reached for the 10-10 second at 120%, the 30-30 second at 110%, and the 15-15 second at 115% of the $\dot{V}O_{2\max}$ (RPE of 10, 10, and 9.9, respectively).

DISCUSSION

The main objective of the present study was to investigate the physiologic responses to the 180° directional changes that occur during intermittent shuttle exercise and that are commonly observed in soccer play. In this context, the comparison of the impact on physiologic responses such as HR, RPE, and [La] between the classical in-line intermittent exercise (IL) and the specific intermittent shuttle exercise with 180° directional changes (IS) was examined. The present study mainly showed that the HR, [La], and RPE were significantly higher when intermittent exercise at the same intensity was performed in shuttle format compared with traditional in-line running.

The HR responses during the 15-15-second and the 10-10-second exercise were similar in both IL and IS forms of running. The directional changes did not affect the central component, in other words, the HR responses. Indeed, in 10-10-second and 15-15-second high-intensity intermittent exercises, the aerobic metabolism contribution was lower than that of the anaerobic system (11) even if these exercise patterns are shown to significantly improve the cardiovascular system (19). A reasonable explanation may be that the work and recovery bouts may not be long enough to attain the slow component (phase III) in the $\dot{V}O_2$ kinetics (12). Indeed, the major part of energy expenditure and substrate use is produced from the adenosine triphosphate/creatine phosphate (PCr) metabolism, glycolysis, and a small part from fat oxidation (20). The passive recovery bouts allowed sufficient reloading of oxygen in myoglobin and hemoglobin, to partially remove a part of [La] produced, and to resynthesize the PCr (12,18). Consequently, the lower participation of the aerobic metabolism and its energy requirement during these efforts could explain why HR responses were not affected in IS with respect to IL.

In contrast with the 10-10-second and 15-15-second exercise bouts, the comparison of %HR_{res} showed greater values in the 30-30-second IS compared with the IL (+7.1%, $p \leq 0.01$). The overall duration of the 10-10-second and the 15-15-second sessions (respectively, 6.50 min and 9.45 min) may be too short to produce significant changes in the cardiac responses to exercise. Bisciotti et al. (6) reported that IL highly taxes the aerobic system. Classically, the 30-30-second IL with active recovery allows athletes to spend a long time at $\dot{V}O_{2\max}$ and therefore improves their $\dot{V}O_{2\max}$ (5), limits glycogen depletion (11), optimizes the use of myoglobin and hemoglobin (28), supports fat oxidation, develops glycolytic enzymes (36), and improves buffer capacity (1). These studies have therefore demonstrated the importance of the aerobic metabolism contribution during 30-30-second IL. However, the application of the 30-30 second in IS format with directional changes would

modify the contribution of the aerobic metabolism to energy expenditure. The central component would appear to be less involved in the energy processes during these particular IS. The 3 180° turn directional changes, which require decelerations and accelerations, relatively break the linear kinetics of the $\dot{V}O_2$ and solicit the anaerobic metabolism at a higher level (32). The importance of the energy expenditure is altered when directional changes are included in intermittent exercises, and, in the case of the 30-30-second exercise, the HR response was higher probably because of the increase in the energy cost and the significant elevation of muscular anaerobic solicitation. These effects were acute and corresponded to the increases in work bout intensity. In this context, Bisciotti et al. (6) has shown that the 30-30-second intermittent in-line protocol at 105% of the $\dot{V}O_{2\max}$ was slightly anaerobic, whereas the same protocol at 110% of $\dot{V}O_{2\max}$ was highly anaerobic in nature.

In relation to IL exercise, RPE and [La] presented higher values for all intensities of IS. These results illustrate the greater physiologic impact generated by the succession of 180° directional changes, which involve decelerations and re-accelerations, as highlighted by Thompson et al. (38) and MacGregor et al. (29). The directional changes in IS require athletes to perform running actions in which they must sharply decelerate and block, implying eccentric muscular efforts and increasing energy cost (32). Moreover, subjects must re-accelerate, and this action solicits essentially the anaerobic metabolism and the fast twitch muscle fibers. Fast twitch fibers possess less mitochondrial and oxidative enzymes than slow twitch, but they present more glycolytic enzymes (PFK, LDH, and MDH) (20). Essen et al. (20) reported that the rate of PCr use decreased by 30% at the end of recovery bouts of a 15-15-second exercise. However, the [La] remained at an identical level to that observed in a continuous 60-minute run. The PCr use decrease could be one of the explanations for the [La] increase observed during the IS, which requires high levels of explosiveness. Another explanation is that glycolytic activity participates in a majority of the energy expenditure during decelerations and accelerations (i.e., intense muscular actions and increased muscular lactacid concentration and thus venous [La]). Consequently, the 2 or 3 directional changes would appear sufficient to fatigue the players more quickly than the IL because a large increase in [La] (ranging from 20.3-25.8%) and RPE (ranging from 25.8-39.6%) was observed. The concentration of PCr would strongly decrease, and the depletion in PCr appears quickly because of the 180° turning movements and the deceleration and re-acceleration phases. The sequence of directional changes combined with high-intensity repetitions (from 100-120% of $\dot{V}O_{2\max}$) and the short duration of the recovery bouts would have increased energy expenditure from the anaerobic metabolism and the energy cost compared with IL. Anaerobic glycolysis would take part more strongly and more quickly in the energy

metabolism, and recovery periods would be too short to sufficiently metabolize [La] and restore PCr.

Furthermore, an active recovery period increases lactate removal after intense exercise in IL (37), whereas the type of recovery does not affect the capacity to metabolize lactate and restore the PCr stock in IS. Indeed, the different intermittent exercises tested in the present study demonstrated similar physiologic responses ([La] and RPE) when they were performed in IS, although the 30-30 second was combined with active recovery and the 10-10 second and 15-15 second with passive recovery. The recovery duration would not be sufficient to highly metabolize lactate and restore the PCr stocks.

The present study has highlighted that the physiologic impact of classical forward short-duration intermittent exercise (IL) was different from that observed in short-duration intermittent shuttle exercise (IS). The IS was shown to increase the intensity of exercise through the additional muscular actions (decelerations and re-accelerations) inducing higher glycolytic contribution, higher [La] values, and higher perceived exertion. The 180° directional changes would imply a faster and more important fatigability throughout the IS in comparison with the IL. However, the 180° turning technique used by some athletes could be better than by other athletes and may have affected results. At high speeds and during high-intensity IS, turning technique becomes more important, and anaerobic power is essential because players must accelerate after turning to obtain the desired speed.

From a methodologic point of view, the Leger-Boucher test (26) might not provide an appropriate $\dot{V}O_{2\max}$ value to perform IS. Indeed, this test is continuous, and it does not consider the directional changes. Coaches and fitness trainers would be able to select the running intensities according to their objectives, but, in IS exercise regimes, these practitioners need a field test that proposes a reference value by taking into consideration the impact of directional changes. Nevertheless, data concerning the effects of directional changes during specific intermittent exercises are not numerous. The 20-m multistage fitness test of Brewer et al. (9) does not provide valid predictions of $\dot{V}O_{2\max}$ (13,31). Consequently, the coach needs a specific value for the intermittent exercises because the $\dot{V}O_{2\max}$ is not precise enough for this type of effort. There may be a notable difference between a $\dot{V}O_{2\max}$ value obtained through a test that involves forward running and a $\dot{V}O_{2\max}$ value obtained in a shuttle run test. This discrepancy has led to the development of the Intermittent Loughborough Shuttle Test (33) and the 30-15 IFT (10). The latter attempts take into consideration the additional energy cost generated by the 180° changes in direction during IS. The 30-15 IFT also improves the prescription of intermittent training interventions either using shuttles or forward runs. Buchheit (10) has shown that the maximal running speed (MRS) determined by the 30-15 IFT presented significantly less variation in

interindividual differences in comparison with the MRS values determined by continuous tests. As the speed increases in the graded shuttle test, the anaerobic metabolism component becomes more important, and, therefore, low individual anaerobic power may cause an athlete to underperform in testing. Finally, the method of Blondel et al. (7) could be more efficient in the application of IS. They showed that, for the same percentage of $\dot{V}O_{2\max}$, the anaerobic contribution to energy supply is different and could be dependent on the individual maximal running velocity ($V_{\max}$). It could be pertinent to perform intermittent exercises in reference to a percentage of the difference between $\dot{V}O_{2\max}$ and $V_{\max}$ (7,16). This suggestion merits investigation, especially in IS running.

PRACTICAL APPLICATION

Previous studies have explored the physiologic impact of classical in-line high-intensity intermittent exercise, but few have analyzed the effects of directional changes in intermittent shuttle exercise that are more specific to soccer play. The results of the present study showed that intermittent shuttle exercise was more intensive than in-line running. Shuttle exercise increased physiologic responses possibly because of the additional muscular actions required in deceleration and re-acceleration actions, inducing a higher glycolytic contribution, higher [La] values, as well as higher perceived exertion. Intermittent exercise with directional changes allows one to solicit different physiologic responses compared with traditional IL and to recreate the common demands of soccer match play. According to the training objectives, the coach can choose between classical in-line intermittent exercise (more aerobic in nature) and specific intermittent exercise, integrating directional changes that promote more soccer-specific activity and a higher anaerobic stimulus.

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